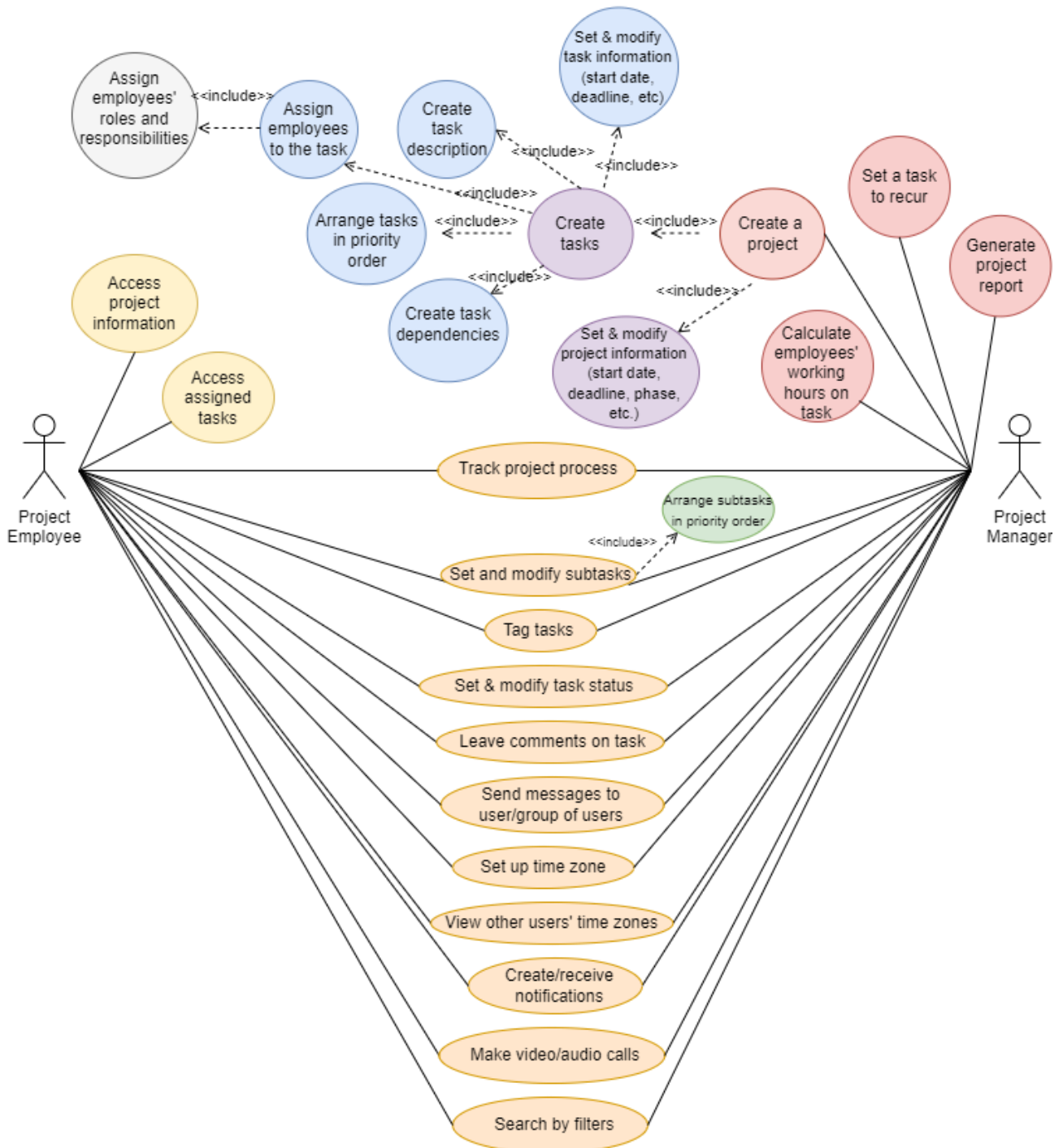


Use Cases

Figure 1

The UML Use Case Diagram of the TMS



Source: Own results.

Use Case Descriptions

Table A1

Use Cases Identified and Initialised from the Project Manager's Perspective

Use Case ID	Use Case Name	Flow of Events	Included Use Cases
UC1	Create a project	1. The project manager decides to create a project. 2. They open the TMS. 3. They select the Projects tab. 4. The system lists all existing projects and provides the Create new project button. 5. The project manager clicks on the Create new project button and starts creating a project.	UC1.1
UC1.1	Set and modify project information	1. The project manager decides to set or modify the project's information. 2. They open the TMS. 3. They select the Projects tab. 4. They select a particular project. 5. The system provides the Edit button next to each project parameter (starting date, deadline, phase, etc.). 6. The project manager clicks on the Edit button and starts either setting new information or modifying the present one.	
UC1.2	Create tasks	1. The project manager decides to create a task for the project. 2. They open the TMS. 3. They select the Projects tab. 4. They select a particular project. 5. The system provides the Create new task button. 6. The project manager clicks on the Create new task button and starts creating a task.	UC1.2.1 UC1.2.2 UC1.2.3 UC1.2.4 UC1.2.5
UC1.2.1	Set and modify task information	1. The project manager decides to set or modify the task's information. 2. They open the TMS. 3. They select the Projects tab.	

		4. They select a particular project. 5. They select a particular task. 5. The system provides the Edit button next to each task parameter (starting date, deadline, status, etc.). 6. The project manager clicks on the Edit button and starts either setting new information or modifying the present one.	
UC1.2.2	Create task description	1. The project manager decides to create the task description. 2. They open the TMS. 3. They select the Projects tab. 4. They select a particular project. 5. They select a particular task. 6. The system provides a textual field to describe the task next to the Description label. 7. The project manager clicks on the textual field and starts describing the task.	
UC1.2.3	Assign employees to the task	1. The project manager decides to assign project employees to the task. 2. They open the TMS. 3. They select the Projects tab. 4. They select a particular project. 5. They select a particular task. 6. The system provides the Assigned to column field to assign project employees to the task. 7. The project manager clicks on the Assigned to column field and mentions (@firstname_lastname) project employees to assign them to work on the task.	
UC1.2.4	Arrange tasks in priority order	1. The project manager decides to arrange project tasks in priority order. 2. They open the TMS. 3. They select the Projects tab. 4. They select a particular project. 5. The system presents a numbered list of the project tasks' titles.	

		7. The project manager draws a task title to the preferred number position in the list.	
UC1.2.5	Create task dependencies	1. The project manager decides to create task dependencies. 2. They open the TMS. 3. They select the Projects tab. 4. They select a particular project. 5. The system presents a numbered list of the project tasks' titles with an empty box next to each title and its number position. 6. The system provides the Set dependency button. 7. The project manager checks empty boxes of two tasks, clicks on the Set dependency button and chooses one of the following options: Finish to Start, Finish to Finish, Start to Start, or Start to Finish.	
UC2	Set a task to recur	1. The project manager decides to set a task to recur at a specified time. 2. They open the TMS. 3. They select the Projects tab. 4. They select a particular project. 5. They select a particular task. 6. The system provides the Recur task button. 7. The project manager clicks on the Recur task button and sets the time and date for the task to recur.	
UC3	Generate a project report	1. The project manager decides to generate a project report. 2. They open the TMS. 3. They select the Projects tab. 4. They select a particular project. 5. The system provides the Generate report button. 6. The project manager clicks on the Generate report button.	

		7. A window to save/print an automatically generated report with the project parameters included pops up.	
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Table A2

Use Cases Identified and Initialised from the Project employee's perspective

Use Case ID	Use Case Name	Flow of Events	Included Use Cases
UC4	Access project information	1. The project employee decides to access the project's information. 2. They open the TMS. 3. They select the Projects tab. 4. They select a particular project. 5. The system presents the project board with all the project parameters described in Section 2.4.1.	
UC5	Access assigned tasks	1. The project employee decides to access the task they have been assigned to work on. 2. They open the TMS. 3. They select the Projects tab. 4. They select a particular project. 5. They select a particular task. 6. The system presents the task board with all the task parameters described in Section 2.4.2.	

Table A3

Use Cases Identified and Initialised from the Project Manager and Project Employee's (Users') Perspectives

Use Case ID	Use Case Name	Flow of Events	Included Use Cases
UC6	Track project process	1. The user decides to track the project's process. 2. They open the TMS 3. They select the Projects tab.	

		4. They select a particular project. 5. The system provides the progress tracking bar, Gantt chart, and Burndown chart. 6. The user views the project's progress on the project board.	
UC7	Set and modify subtasks	1. The user decides to set or modify the task's subtask. 2. They open the TMS. 3. They select the Projects tab. 4. They select a particular project. 5. They select a particular task. 6. The system provides the Create new subtask button in the to-do list field. 7. The system provides the Edit button next to each created subtask. 8. The user clicks on the Create new subtask button and starts devising a subtask. 9. The user clicks on the Edit button and starts editing a subtask.	UC7.1
UC7.1	Arrange subtasks in priority order	1. The user decides to arrange the subtasks in priority order. 2. They open the TMS. 3. They select the Projects tab. 4. They select a particular project. 5. They select a particular task. 6. The system presents a numbered list of the task's subtasks. 7. The user draws a subtask to the preferred number position in the list.	
UC8	Tag tasks	1. The user decides to tag the tasks. 2. They open the TMS. 3. They select the Projects tab. 4. They select a particular project. 5. They select a particular task. 6. The system provides the Tags field to tag the task.	

		7. The user clicks on the Tags field and tags (#tasktag) the task.	
UC9	Set and modify task status	1. The user decides to set or modify the task's status. 2. They open the TMS. 3. They select the Projects tab. 4. They select a particular project. 5. They select a particular task. 6. The system provides the Edit button next to the Status field. 7. The user clicks on the Edit button and selects one of the task status options: Awaiting Assignment, In Progress, In Review, Completed or To Be Modified.	
UC10	Leave comments on a task	1. The user decides to leave a comment on the task. 2. They open the TMS. 3. They select the Projects tab. 4. They select a particular project. 5. They select a particular task. 6. The system provides the Comments field where the user can view other users' comments on the task and write a comment themselves. 7. The user writes a comment in the Comments field and posts it by clicking on the Post button.	
UC11	Send messages to user/group of users	1. The user decides to send a message to another user or a group of users. 2. They open the TMS 3. They select the Messages tab. 4. The system lists all the user's conversations that have taken place. 5. The user clicks on the New message icon. 6. The system presents the Search employees field. 7. The user types the name and/or last name of the user they are willing to converse with. If the user is willing to converse with a group of users, they click on the Create a group button and type their names and/or last names.	

		8. The system lists all the users with such names and/or last names. 9. The user clicks on the specified user(s). 10. The system provides a window where a conversation is held. 11. The user writes a message and clicks on the Send button.	
UC12	Set up time zone	1. The user decides to set up their time zone. 2. They open the TMS 3. They select the Settings tab. 4. They select the Time zone icon. 5. The system provides the default time zone for the company's default shift hours. 6. If the user is willing to set up the default time zone, they click on the Set up as default button. If the user is willing to set up a time zone they are currently in, they click on the Choose time zone button, the system provides a list of time zones, and the user clicks on a particular one.	
UC13	View other users' time zones	1. The user decides to view another user's time zone. 2. They open the TMS. 3. They hover their mouse cursor over the other user's name and view their time zone.	
UC14	Create/receive notifications	1. The user decides to create a notification. 2. They open the TMS. 3. They select the Settings tab. 4. They select the Notifications icon. 5. The system provides the Create notification button. 6. The user clicks on the Create a notification button. 7. The system provides the Set time button. 8. The user sets the exact time for a notification to appear.	

		9. The system provides the Select type button with two options to select from: sticky notifications or full-screen notifications. 10. The user clicks on the Choose type button and chooses their notification type. 11. The system provides a textual field to describe a notification. 13. The user writes a description of their notification. 14. The user clicks on the Save button.	
UC15	Make video/audio calls	1. The user decides to make a video/audio call. 2. They open the TMS. 3. They hover their mouse cursor over the user's name. 4. The system provides the Make video call and Make audio call icons. 5. The user clicks on either of them and the integrated application Zoom starts the call.	
UC16	Search by filters	1. The user decides to search for a task or project by filters. 2. They open the TMS. 3. They click on the Search tab. 4. They click on the Search by filter icon. 5. The system lists the filters by which the search can be carried out: the project's title, starting date, deadline, status, condition, phase and the task's title, starting date, deadline, project employees, status, and tags. 6. The user selects one or more filters. 7. The system lists the tasks or projects that meet the selected filters.	